

Riccardo Pazzi

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Research statement

Passionate about cryptography, ML and its social impact. Interested in theoretical aspects, mainly around issues such as the robustness of algorithms and privacy/security. Gained practical experience by developing several projects related to privacy preserving technologies, NLP and Machine Learning.

Education

PHD COMPUTER SCIENCE

BIELEFELD UNIVERSITY, 2026-2029

Topic: Physics informed neural networks and adversarial robustness in Machine Learning for critical infrastructures

MSC COMPUTER ENGINEERING

POLITECNICO DI MILANO, 2024

GRADE: 110/110

Relevant courses: Machine Learning, Artificial neural networks and Deep Learning, Recommender Systems, Quantum computing (Ph.D. course), Computing infrastructures

MSC DATA SCIENCE

POLYTECH NICE-SOPHIA, 2023

HIGHEST HONOURS

Relevant courses: Privacy and Blockchain, Advanced Deep Learning, Data mining on graphs, ML and image processing, Graph algorithms and optimization

EIT DIGITAL PROGRAM IN DATA SCIENCE

Double degree from two European institutions, with a **major in Data Science** and a **minor in innovation and entrepreneurship**.

BSC COMPUTER SCIENCE AND ENGINEERING

POLITECNICO DI MILANO

GRADE: 98/110

Relevant courses: Probability and Statistics, Linear algebra and geometry, Analysis I and II, Logic and Algebra, Algorithms and theoretical computer science, Operating systems and computing architectures.

Publications

HEDEL: HOMOMORPHICALLY ENCRYPTED DISTRIBUTED ENSEMBLE LEARNING

PAZZI RICCARDO, FALCETTA ALESSANDRO, ROVERI MANUEL: IJCNN, 2025

Combines **distributed multi-key homomorphic encryption with transfer learning** to securely merge models from different providers: In settings such as healthcare this approach could substitute classical Federated Learning approaches reducing communication between parties and improving security guarantees.

Work Experience

PRIVACY PRESERVING MACHINE LEARNING ENGINEER

DHIRIA SRL, 2024-2026

- Developed **HEFrame**, a library which provides higher level ML operations using homomorphic encryption primitives, with a focus on efficiency.
- Developed a **solution for encrypted text classification** aimed at recognizing sensitive documents without exposing the content.
- Gained expertise in NLP, privacy preserving techniques and MLOps.

RESEARCH INTERNSHIP

CNRS/UNIVERSITÉ COTE D'AZUR, 2022

- Investigated optimal policies for entanglement swapping in a quantum networks. Policies were computed using Deep Q-learning.
- Reviewed state of the art in entanglement swapping

ARTIFICIAL INTELLIGENCE FOR CYBERSECURITY

UNIVERSITY OF OXFORD, 2024

The course + project covered the technical and business aspects of implementing artificial intelligence solutions in the cybersecurity domain

TEACHING TUTOR

POLITECNICO DI MILANO

Teaching tutor for the course **Probability and statistics** (Undergraduate)

Awards

EIT DIGITAL MASTER SCHOOL SCHOLARSHIP

The scholarship completely covered the costs for 2-year MSc in Data science (2000€/year)

FULL REIMBURSEMENT FOR PETS 2023

Received a full reimbursement (travel + housing) as a master student to attend the PETS (Privacy Enhancing Technologies Symposium) conference in Lausanne in 2023

Activities/Interests

ON AIR, THE VOICE OF SCIENCE

OnAir is an organization created by university students, the goal is to **reach people outside the university bubble** through talks, labs, and other events dedicated to scientific dissemination.

We raised >10k € for science dissemination activities and collaborate with important national theatres.

VOLUNTEER FOR LEGAMBIENTE

Legambiente is an Italian association dedicated to environmental issues and endangered species monitoring.

CERTIFIED SCUBA DIVER (P3 FIPSAS/CMAS)

EXTRAS: TEA, TURTLES AND PHILOSOPHY

I like drinking tea or coffee while reading good philosophy, lately I've been engaged with moral and political philosophy, but I'm also interested in philosophy of science and philosophy of mind. Finally I have a collection of turtle statuettes in my bedroom and I try to buy a new one for every country I visit, right now I'm sitting at 14.